Handout 11: The perft reference table

`perft(depth)` counts every position the legal move generator can reach in exactly `depth` plies. The counts below are the agreed answers for five standard positions and for the warm-up position. A generator that prints the same numbers is right in every rule those positions use: castling, en passant, promotion, pins, and checks.

Position	FEN	Depth 1	Depth 2	Depth 3	Depth 4
Start	`rnbqkbnr/pppppppp/8/8/8/8/PPPPPPPP/RNBQKBNR w KQkq - 0 1`	20	400	8,902	197,281
Kiwipete	`r3k2r/p1ppqpb1/bn2pnp1/3PN3/1p2P3/2N2Q1p/PPPBBPPP/R3K2R w KQkq - 0 1`	48	2,039	97,862	
Position 3	`8/2p5/3p4/KP5r/1R3p1k/8/4P1P1/8 w - - 0 1`	14	191	2,812	43,238
Position 4	`r3k2r/Pppp1ppp/1b3nbN/nP6/BBP1P3/q4N2/Pp1P2PP/R2Q1RK1 w kq - 0 1`	6	264	9,467	
Position 5	`rnbq1k1r/pp1Pbppp/2p5/8/2B5/8/PPP1NnPP/RNBQK2R w KQ - 1 8`	44	1,486	62,379	
King and pawn against king (handout 5)	`4k3/8/8/8/8/8/4P3/4K3 w - - 0 1`	6	30	210	1,424

Kiwipete holds every special move at once: both sides can castle, a pawn can be promoted, and an en passant capture is possible. When a count is off, run `divide()` on the same position: it prints the count under each root move, so the one move whose count differs from the reference points at the bug.

Remus in plain Python counts a few thousand positions a second, so depth 3 from the start finishes in seconds and depth 4 takes about a minute. The blank cells are counts that take too long for a class period.